

# TOLUPE MATTHEW BUSOYE

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**Research Interests:** TinyML & Edge AI · Embedded Systems · Cyber-Physical Systems · Intelligent Energy Systems · Internet of Things · Renewable Energy

## Education

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**Master of Science (M.Sc.) in Mathematical Sciences (AI for Science Stream)** 2026 – 2027  
*African Institute for Mathematical Sciences (AIMS) South Africa / Google DeepMind*

- Theoretical knowledge and practical skills in mathematics, machine learning, and AI for scientific applications.

**B.Sc. Electrical and Electronic Engineering** 2019 – 2025  
*University of Ibadan, Nigeria* CGPA: 3.94/4.00 — First Class

- Best Graduating Student of the Department and Faculty of Technology
- Studied electronic circuits, microprocessors and embedded systems, electric machines, programming, and cross-disciplinary engineering courses
- **Thesis:** Development of a Commercialised Solar-Based Mobile Energy Hub

**Senior Secondary Certificate Examination** 2015 – 2017  
*Pacesetters Comprehensive College*

- Ranked among the top 5%; received the best student award in Physics, Mathematics, and Economics

## Research Experience

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**TinyML & Edge AI Lead**, ML Collective Apr 2026 – Present  
Remote. Mentored by Steven Kolawole and independent researchers within the ML Collective network. Model optimisation and deployment on microcontrollers for real-world applications; focus on energy-efficient embedded inference.

**Hardware & TinyML Researcher**, Powerstart Integrated Technologies (PowerLabs) 2025 – 2026  
Lagos, Nigeria. Supervised by Dr. Eigbe Egbonbon-aye. Optimised TinyML models for deployment on microcontrollers and embedded systems with as little as 2kB of available memory. Developed deep microcompression pipeline combining structured pruning and bit-packed quantisation; work accepted to ICML 2026 Global South in AI Workshop.

**Lead Student Investigator**, University of Ibadan 2024 – 2025  
Ibadan, Nigeria. Supervised by Prof. O. O. Olakanmi. Developed a commercialised solar-based mobile energy hub for off-grid communities. Designed full-system hardware: circuit schematics, PCB layout, firmware in C/C++, and mobile application in Flutter/Dart.

## Publications and Articles

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### Conference Papers

- T. M. Busoye, Q. A. Jokomba, O. M. Busoye, A. O. Joleto. **Edge-Deployed Machine Learning Sensor (MLSensor) for Real-Time Anomaly Detection in Electrical Submersible Systems.** *SPE Nigeria Annual International Conference and Exhibition, 2026.* DOI: [10.2118/235027-MS](https://doi.org/10.2118/235027-MS)
- O. M. Busoye, T. M. Busoye, E. Eigbe. **Deep Microcompression: Structured Pruning and Bit-packed Quantization for Microcontrollers.** *PMLR Vol. 306 — ICML 2026 Global South in AI Workshop.* DOI: [10.48550/arXiv.2609.05081](https://doi.org/10.48550/arXiv.2609.05081)

### Manuscripts Under Review / In Preparation

- O. M. Busoye, T. M. Busoye, E. Eigbe. (in preparation). **Right In-Place Convolution: A Simple, General, and Near-Optimal Strategy for Memory-Efficient CNN Inference.**
- T. M. Busoye. (in preparation). **A Compile-Time and Memory-Efficient Approach to SunSpec Models with FlatBuffers and Direct Modbus Register Mapping.**

### Thesis & Articles

- T. M. Busoye. *Development of a Commercialised Solar-Based Mobile Energy Hub.* Undergraduate thesis, University of Ibadan, 2025. DOI: [10.13140/RG.2.2.10441.86886](https://doi.org/10.13140/RG.2.2.10441.86886)
- T. M. Busoye. Inside TensorFlow Lite Micro on RP2040/RP2350: Setup and Project Structure Explained. *Medium*, Sep 2025. [Link](#)

## Industrial Experience

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**Embedded Systems Engineer** Jun 2025 – Present  
*Powerstart Integrated Technologies Limited (PowerLabs), Lagos*

- Led the development of the emobility charging station for energy trading.
- Research and evaluate sensors for integration with Powerstart Autonomous Infrastructure (PAI) devices
- Implement SunSpec protocol to enable communication with SunSpec-compliant equipment
- Develop a hardware testing framework and integration-testing workflow supporting test-driven development
- Create and maintain a coding style guide to standardise hardware-related software development

#### **Embedded Systems Intern**

*Jan 2024 – Jun 2024*

*Powerstart Integrated Technologies Limited (PowerLabs), Lagos*

- Designed and developed a PAI device for power event capture based on Raspberry Pi Pico MCU
- Collaborated on software components for transmitting power events to Amazon AWS cloud endpoints
- Designed schematics and PCBs for Power Stool, a Bluetooth-controllable desk

#### **Hardware & Electronics Intern**

*Jun 2023 – Sep 2023*

*Reeddi Technologies, Lagos*

- Repaired 50 Reeddi Capsules, Energy Boxes, and Big Energy devices
- Designed a charge station for 100 Reeddi Capsules and set up Big Energy installations
- Developed device tracking systems

#### **STEM Tutor & Engineer**

*2021 – 2024*

*Switches Technology, University of Ibadan*

- Tutored and trained over 10 students in Arduino, Raspberry Pi, and embedded systems
- Designed innovative projects in embedded systems, IoT, and AI—including a Smart Distribution Board, Smart Bin, and Responsive Prosthetic Leg
- Oversaw 3D printing projects including a Head Mount Mouse and a 6-DOF Robotic Arm

#### **Electrical & Electronic Engineering Intern**

*Apr 2022 – May 2022*

*Crittall-Hope Nigeria Limited*

- Maintained single and three-phase machines; contributed to roller shutter production

#### **Electrical & Electronic Engineering Intern**

*Aug 2020 – Mar 2021*

*God's Will Technology Concept*

- Installed solar and inverter systems; performed electrical installations and electronic repairs
- Received foundational training in electrical systems and electronic repair techniques

## **Teaching Experience**

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#### **Academic Tutor**

*2021 – 2023*

*Faculty of Technology, University of Ibadan*

- Tutored over 70 fellow students in multiple courses including Elementary Computer Programming (FORTRAN)
- Tutored over 100 first-year students in PHY103 and MAT121

#### **Academic Tutor**

*2021 – 2022*

*Toppers Tutorial, University of Ibadan*

- Tutored over 100 first-year students in PHY103 and MAT111

#### **Academic Tutor**

*2021 – 2025*

*Mountain of Fire and Miracle Ministries Campus Fellowship, UI*

- Prepared academic materials and organised tutorials for colleagues and junior students

## **Leadership and Volunteering**

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#### **Lead, ML Systems Study group**

*2026*

*ML Collective*

- Managed over 20 students in to understand the foundational principles of Machine learning systems using the cs249r\_book by Harvard University.
- Led weekly study sessions where we discussed about each week's chapter and programming module implementation.

#### **President, Society of Electrical & Electronic Engineering Students (SEEES)**

*2023/2024*

*University of Ibadan*

- Responsible for over 270 electrical and electronic engineering students
- Organised an industrial visit to Egbin Power Plc, Nigeria's largest power-generating station
- Coordinated workshops bridging theoretical knowledge and industry-required practical skills
- Increased financial resources for the next administration by over 2,000%

**Lead & Co-Lead, Hardware & IoT, Google Developer Student Club (GDSC)**  
*University of Ibadan*

2021/22 – 2022/23

- Organised technical sessions and hardware workshops for over 50 students
- Served as Technical Lead, TechQuest GDSC UI 2023

**Public Relations Officer, SEEES**  
*University of Ibadan*

2021 – 2023

- Led publicity and communication for all society activities; served as official spokesperson

**Student Representative Council Member, 10th Assembly**  
*Student Union, University of Ibadan*

2022/2023

- Represented 100 students in constituency; responsible for policy-making
- Member, UISU House Committee on Audit and Electoral Matters

## Scholarships and Awards

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- **3rd Place—Africa Energy Challenge, SPE Africa Annual Student Congress** Apr 2025  
Pan-African student energy-innovation competition hosted by the Society of Petroleum Engineers.
- **1st Place—SPE Energy Challenge, Student Technical Symposium & Exhibition, Nigeria** Mar 2024  
National student energy-engineering competition among universities by the Society of Petroleum Engineers, Nigeria.
- **NNPC/TotalEnergies National Merit Scholarship** 2021 – 2024  
Competitive national scholarship for top-performing Nigerian STEM undergraduates.
- **Federal Scholarship Board (FSB), Nigeria** 2021 – 2023  
Nigerian federal government merit scholarship for outstanding undergraduates.
- **SPE Lagos Section Male Undergraduate Scholarship (Engineering)** 2023  
Merit scholarship awarded by the SPE Lagos Section to top engineering undergraduates.
- **Agbami Medical and Engineering Professionals Scholarship Award** 2019 – 2022  
Merit-based scholarship for Nigerian engineering and medical students.
- **Most Outstanding in Academics—University of Ibadan Student Union** 2023/24  
Top academic distinction among the University's undergraduate student body.
- **5× Dean's Honour List—Faculty of Technology** 2018/19 – 2023/24  
Awarded each session to the top-performing students in the Faculty.
- **Most Outstanding Member—Junior Chamber International Nigeria, UI** 2022/24  
Recognises leadership and service contribution within the chapter.
- **4× Academic Excellence Awards—MFMCF UI** 2020/21 – 2023/24  
Annual academic distinction awarded within the fellowship.
- **Silver Medal (1500m), Bronze Medal (400m)—TESA Dean's Cup** 2021/22  
Inter-departmental athletics competition, Faculty of Technology.

## Workshops and Conferences

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- Adapting to Change, Edinburgh Earth Initiative, University of Edinburgh Nov 2024
- Nigeria Annual International Conference and Exhibition, SPE Nigeria Aug 2025
- SPE Africa Annual Student Congress, Society of Petroleum Engineers, Africa Apr 2025
- Student Technical Symposium & Exhibition, SPE Nigeria Mar 2024
- Student Professional Awareness Conference, IEEE, University of Ibadan Jul 2023
- Exhibitor, Energy Searchlight 2.0, Energy Club, University of Ibadan Feb 2025
- Exhibitor, Ambizero Conference, Energy Club & SPE, University of Ibadan Jun 2024
- Artificial Intelligence and Clean Energy (AI4CE) Workshop, First Technical University 2021

## Selected Projects

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- **IoT Air Quality Monitoring System** — End-to-end real-time monitoring solution with cloud storage and mobile/desktop dashboard; awarded £1,500 grant by the Edinburgh Earth Initiative. [GitHub](#)
- **Edge-Based Smart Meter** — Offline metering system for local energy data acquisition and TinyML-based analytics on constrained embedded hardware. [GitHub](#)
- **Power Leaf** — Pay-per-use solar energy platform integrating IoT monitoring, prepaid billing, and mobile control for scalable renewable-energy access. [GitHub](#)
- **Automated Solar Power Trading and Distribution Unit** — Smart solar-power distribution prototype with priority-based energy arbitration; 1st place, SPE Energy Challenge 2024
- **Hybrid Solar/Grid Charging Station for 100 Reeddi Capsules** — Embedded power-system design for distributed charging and energy management in a battery-swapping ecosystem
- **STM32 PCB Design and Fabrication · PCB Design for Smart Desk & Power Stool · Automatic Solar Powered Traffic Light (RAINigeria)**
- **Smart Energy Monitoring and Distribution Board · Bluetooth Programmable Display (Arduino + Flutter) · Responsive Prosthetic Leg**
- Over 15 solar and inverter installations across residential and commercial sites

## Skills and Languages

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**Technical:** Circuit design, PCB design, Solar and inverter installation, Electronic prototyping  
**Software:** C/C++, Python, Zephyr, Linux, Buildroot, Pytorch, KiCad, Docker, Git, Flutter/Dart, L<sup>A</sup>T<sub>E</sub>X, Microsoft Office, Canva  
**Strengths:** Fast learning, Communication, Leadership, Time management, Adaptability, Creativity  
**Languages:** English (fluent), Yoruba (native)

## Professional Affiliations

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- Graduate Member, Nigerian Society of Engineers (GMNSE)
- Student Member, Society of Petroleum Engineers (SPE)
- Student Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, Society of Electrical and Electronic Engineering Students (SEEES), UI
- Member, Energy Club, University of Ibadan
- Student Member, Google Developer Student Club (GDSC)
- Member, Junior Chamber International Nigeria (JCIN)